What topology hold-out removes—and what it does not

Running title: What does topology hold-out actually test?

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Abstract

Topology hold-out evaluates metal–organic frameworks with topology labels absent from training, but coupled chemical support limits interpretation of its score; we introduce a metadata-based audit of exact composition-key support, conditional entropy and pairwise overlap. In hMOF (324,426 structures), 74.7% of coarse composition keys occur in one topology and account for 61.3% of structures, while seven of eleven topologies lose all same-key training counterparts but contain only 21.3% of structures; ARC-MOF provides a partly dependent audit under different keys. For CO_2 uptake at 0.15 bar and 298 K on the sole usable matched pair, pcu ↔ nbo, partner removal increased nbo-side MAE by 0.0746 mmol/g (paired 95% t interval 0.0591–0.0900; 70.0% of the reference mean), whereas the pcu-side change was not detected and size-matched random thinning did not reproduce the nbo degradation. Across five 25,000-row redraws, retrospective out-of-fold top-1% recovery decreased from 50.4% under random splitting to 19.3% under topology grouping (paired difference 31.1 percentage points, 95% t interval 24.6–37.6); representation, replacement and fixed-pool follow-ups reveal conditional performance effects, supporting disclosure of chemical coverage alongside grouped scores without identifying a chemical mechanism, a calibrated support-to-error relation or external predictive validity.

Keywords: metal–organic frameworks; CO_2 capture; machine learning; data leakage; confounding; topology hold-out; grouped cross-validation

1. Introduction

1.1 The protocol that decides the claim

Machine-learned screens of MOFs report high random-split R^2 for CO_2 uptake — 0.93–0.97 at 0.15–2.5 bar in ref. 17 — and for other MOF properties (ref. 26), and such numbers are used in materials-selection workflows for post-combustion CO_2 capture at 0.15 bar and 298 K (refs. 2, 4); under a topology hold-out the same class of model spans R^2 below 0.10 to ≥ 0.97 depending on the target and the pressure range (ref. 8). The dominant hMOF database is **enumerated**: a finite set of building blocks is recombined on a library of topological templates (ref. 1), so many entries are close relatives and a random split places relatives on both sides of the train/test boundary. ARC-MOF is a mixed-source library (refs. 3, 15) that partly overlaps hMOF (§3.5). The result is a familiar but under-specified situation: the number a screen reports depends on who ends up on which side of the boundary, and the boundary is set by the protocol, not by the data.

Topology hold-out evaluates prediction on unseen topology groups. When chemistry is strongly coupled to topology, that task also entails chemical-support shift; its score alone cannot isolate an effect attributable to topology. This work quantifies the support the split removes and examines the consequences through controlled training-pool contrasts and screening metrics.

1.2 What prior work establishes, and what this work adds

The shift from reporting a score to reporting the protocol that produced it is recent and well documented. Three 2026 studies are the clearest points of reference, and each overlaps this work in a specific way. Asgari et al. held the model, descriptors, hyper-parameters and metric fixed and varied only the split, finding that the split itself defines the discovery claim — under their design R^2 falls from 0.758 under random splitting to 0.267 under topology grouping, and top-5% recovery falls from 62.2% to 37.6% (ref. 16). Alimardani et al. extended that result across a common cohort of 255,939 MOFs and 27 adsorption targets under five non-diagnostic descriptor families, showing that descriptor adequacy is regime-dependent — for methane uptake at 5.8 bar, geometry-only R^2 is ≈ 0.30 against ≈ 0.69 for geometry plus RAC chemistry (ref. 8). Wang et al., on the same database and the same 0.15 bar target used here, found by ablation that chemistry-type descriptors contribute most of the gain beyond geometry (ref. 17).

These studies are careful about what they read a topology hold-out as. Alimardani et al. interpret it "as a deployment proxy for structural extrapolation" while stressing that it "does not imply that topology labels are the mechanistic cause of prediction error" (ref. 8). Asgari et al. read topology- and node-family hold-outs as regimes that "combine chemical-family transfer with descriptor-space shift", describing the topology-grouped penalty as "a combined stress test involving topology-label granularity, group-size imbalance, descriptor coverage, descriptor-space shift, and target sensitivity" rather than a statement that topology alone mechanistically determines adsorption (ref. 16). Related analyses by the same research group go further in the same direction: they build chemically auditable descriptor atlases (ref. 9), treat benchmark maturity as a precondition for a reliable conclusion (ref. 10), and ask when topology matters beyond pore metrics (ref. 28). Prior work therefore already treats the protocol as a mixed stress test rather than a clean separation of topology from chemistry, and this paper claims no priority on either finding.

This work measures the size of the mixing. To our knowledge, the three studies compared above do not report the composition-to-topology determinism — whether as the single-topology composition fraction or as H(topology | chemistry) — nor do they connect such a number to the support a topology hold-out actually removes from training. Related work shows topology is very unevenly distributed across MOF databases (ref. 7), that MOF data carry known biases and redundancy (ref. 6), and that identifier schemes such as MOFkey exist precisely to detect structural duplicates (ref. 27); these are descriptions of the data, not of the hold-out. A grouped split remains a legitimate deployment proxy even when the blocking variable is confounded with the predictors; the prior-art point (refs. 18–20) is only that the confound changes what the score can be read as, which is why the chemical support must be disclosed alongside the score. The same tension is documented outside MOFs: scaffold and time splits are standard in molecular benchmarks because random splits overstate prospective accuracy (refs. 21–23), and the graph-based model classes now common in this literature (ref. 24) are trained and evaluated under the same split definitions, so the choice of split bears on them as well. Molecular out-of-distribution benchmarks, QSAR partitioning studies and chronological evaluations all make the split itself the variable under test (refs. 29–31). What this work adds is the measured magnitude of the coupling — in hMOF (an enumerated library) and in ARC-MOF (a mixed-source library under its own metadata definitions) — and, in hMOF, of its consequence; whether the same magnitude holds for enumerated libraries generally is not tested here. Crucially, a measured support loss is not a measured error attribution: the counts fix the identity of the withheld chemistry, not the cost of withholding it.

1.3 Contributions

Nothing here claims the phenomenon: a topology hold-out mixes structural and chemical shift, and refs. 8 and 16 already read it that way. The claim is the size of the mixing and the instrument that reports it. Three contributions follow. (i) A support audit needing only a composition key and a group label — per-group unseen fraction, the conditional-entropy ratio in both directions, and pairwise composition overlap — implemented as a diagnostic that runs on a table the reporter already holds, with input checks for the identifiers it needs; it is paired with a 22-rule reporting checklist and companion tool (SI-A.3) that state what must accompany a grouped-split score and what such counts cannot support. (ii) Fixed-test-set contrasts on the one usable chemistry-matched topology pair distinguish partner removal from random thinning; follow-up replacement, representation and novelty analyses test the limits of interpreting those contrasts. (iii) Retrospective out-of-fold and fixed-candidate-pool screening analyses distinguish validation rankings from the output of an individual fitted model. All of it concerns validation scope within the studied data, rather than an identified adsorption mechanism or an externally validated screening method.

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2. Methods

2.1 Data, sampling and composition definitions

The primary database is the hMOF release (ref. 2) — the enumerated library built with the procedure of ref. 1 — comprising 324,426 structures. That count is the row count of the file `all_MOFs_screening_data.csv` in the Materials Cloud record `nsp7v-2dk91` cited under Data availability, not a value printed in Boyd et al. (ref. 2): ref. 1 (Wilmer et al.) supplies the enumeration procedure and ref. 2 (Boyd et al.) the release that this record distributes, so a reader who wants this exact library must take the archived file rather than the article text. The hMOF release contains 42 fields; models use eight numerical structure descriptors and the categorical feature blocks specified in §2.3 (the SI parameter table lists the field groups). The elite release subset is 8,206 structures. The second database is ARC-MOF v6 (ref. 15), 264,225 structures, from which only the topology and chemical-cluster metadata are used, because it carries no adsorption property in a form this study's models consume, so it cannot serve as an external prediction test. Both are public; neither is redistributed (Data availability). Modelling samples of 25,000, 50,000 or 100,000 rows are drawn with a fixed seed unless a multi-seed design states otherwise: the 25,000-sample scale is the screening experiment of §3.4, the 50,000-sample scale is the controlled intervention of §3.3 (large enough that the pcu ↔ nbo shared compositions carry usable row counts, small enough to keep the many-arm refit cheap), and the 100,000-sample scale is the protocol comparison of §3.2. The full-database support figures use all rows and are always reported with their n, because the single-topology fraction is sample-size dependent: it reads 98.9% at n = 25,000 against 93.2% at n = 324,426 under the fine definition, so a number without n is not comparable to anything.

Chemical composition has two granularities. The **coarse** definition — the headline — concatenates the database metal-node identifier and the two role-specific organic-linker identifiers (the `metal_linker`, `organic_linker1` and `organic_linker2` fields); the **fine** definition adds the functional group (`functional_groups`). A structure whose key has no exact counterpart in training is **unseen** under that definition. The two databases define "a composition" with different keys: hMOF uses building-block identities, whereas ARC-MOF uses its own metal/linker/functional cluster identifiers, so the two are not interchangeable and cross-database magnitudes are never ranked against each other. For hMOF, the 1,997 structures with an unannotated functional group are treated as one additional fine-grained value rather than dropped, so that the fine definition does not silently discard rows.

2.2 Support, entropy and overlap measures

Box. Reading guide for non-specialists. *Topology hold-out* means: train on some framework types (nets) and predict on a net the model has never seen. Because chemistry is tied to the net, this also withholds chemical relatives, so the score cannot say how much of any penalty is structural and how much is chemical. A *composition key* is the set of building-block identities (metal node + two linkers, optionally + functional group) that labels a structure's chemistry. *GroupKFold* splits the data into five folds by a grouping label so no group straddles train and test; *leave-one-topology-out* (LOTO) is the extreme case that holds out one topology at a time and trains one model per held-out topology — the two are different protocols, not two precisions of one. The support audit reports three counts: the share of chemistries private to one topology (composition-weighted), the share of structures that lose their same-key training match (structure-weighted), and an entropy ratio H(T|C)/H(T) measuring how far knowing the chemistry narrows down the topology (0 = chemistry pins the topology completely; 1 = no narrowing at all). *Jaccard* overlap is shared keys divided by the union, used to ask whether two topologies share any chemistry at all. *Miller–Madow* is a first-order correction for the downward bias of entropy estimates from finite counts. *SHAP* attributes a fitted model's prediction to individual features. *Paired t-intervals* on five seeds describe seed-to-seed stability, not uncertainty over new materials.

Three quantities describe the support a hold-out removes. (i) The **single-topology fraction** — the share of distinct compositions that occur in exactly one topology — weights compositions equally and answers "how much of the chemistry space is private to one topology?". (ii) The **structure-level unseen fraction** — the share of structures whose key has no training counterpart after their topology is held out — weights structures and answers "how much of the database loses its same-key counterpart?". The two differ whenever common compositions are more populous, so both are always reported. (iii) The **conditional-entropy ratio** H(topology | chemistry) / H(topology) is structure-weighted because the deployment question is about structures, not composition labels: a composition that appears 10,000 times should weigh 10,000 times more in the entropy than one that appears once, and the structure weighting makes the ratio answer "how much of the structure mass does conditioning on chemistry resolve?". The relation is asymmetric, so the reverse direction is also reported: H(chemistry | topology) / H(chemistry) = 0.860 means topology removes only 14% of composition uncertainty. Quoting 18.0% alone would invite the false reading that the two variables mutually determine each other.

The plug-in entropy carries a first-order Miller–Madow correction $\Delta H(T|C) = (K_{TC} - K_C)/(2\,N \ln 2)$, where K_C and K_{TC} count occupied marginal and joint cells. For the sparse fine-granularity space this correction is substantial (≈ 22% of the estimate, 0.0307 bit against $H(T|C) = 0.139$ bit) and is reported as a sensitivity check rather than a proof of unbiasedness, because a space with many singletons need not have its bias removed by a first-order term; the coarse correction is negligible (0.000912 bit). The pairwise-overlap profile completes the audit: it lists, for every topology pair, the count of compositions they share and the Jaccard index (the shared count over the union of the two composition sets), which determines whether the exact-composition-matched contrast used here has shared-key support. A pair with zero exact-key overlap cannot support that contrast.

2.3 Models, features and validation protocols

The original protocol and intervention analyses use a random forest (120 estimators by default; the ten-target panel, five-seed 25,000-sample LOTO runs and S_g curve use 100, as specified in the SI parameter table). Models and features are held fixed within each protocol contrast. The choice permits repeated fits and the SHAP diagnostics (ref. 25); it is not a claim of optimal model performance. The original 426-column representation comprises eight structure-derived numerical descriptors plus one-hot blocks for topology (11 levels), metal-node identity (7) and functional groups (400). Linker identifiers are grouping variables and omitted-variable diagnostics in these original analyses, and become model inputs only in the follow-up comparisons (§2.7). A 415-column check removes the topology block from both protocol arms. This controls access to that block but does not establish adequate chemical representation. Gradient boosting and ridge (refs. 32, 33; implemented with refs. 11–14) provide additional within-family protocol comparisons from one cross-validation run per protocol, without seed intervals or an outer hyperparameter search; ridge selects its regularisation parameter internally by cross-validation. The follow-up representation study adds ExtraTrees; its encoding and feature-subsampling settings differ from the original analyses and its absolute scores are therefore reported separately.

Two grouped-validation constructions appear and must not be conflated. A **5-fold GroupKFold** by topology (the §3.2 baseline) gives a pooled R^2 of 0.3786. A **leave-one-topology-out** (LOTO) scheme, scored once per held-out topology, gives the four aggregation rules compared in §3.2. The paired protocol comparisons of this study are designed so that two arms share the same test target values and total sum of squares; their prediction-error sums of squares then differ, which is exactly the quantity the comparison measures — a shared denominator is not a shared numerator. R^2 across different test populations can still be reported, but such a difference cannot be attributed solely to the model or protocol, because the denominator differs too.

2.4 Fixed-test-set interventions (arms a–f, post hoc arm g)

The controlled contrast uses 50,000 samples and the single chemistry-matched pair pcu ↔ nbo (408 shared compositions). For each of five independent seeds, the test set is drawn once and held fixed; only the training pool changes across arms: **(a)** with the partner topology included; **(b)** with the partner topology removed; **(c)** a size-matched random draw from (a) down to (b)'s training size; **(d)** a draw matching (b)'s composition/topology profile; **(e)** removing only those partner rows whose composition also occurs in the test set; **(f)** a size-matched random draw from (e). A seventh arm, **(g)**, is post hoc rather than prespecified: from (a) it removes the same number of partner rows that (e) removes, drawn at random without conditioning on whether their composition recurs in the test set (SI-C.4). It was run after the a–f design was fixed and is used only as a control in §3.3. The seed-fixed test set is what makes R^2, MAE and RMSE directly comparable across arms, because the target variance of the test set does not change between arms — the only thing that changes is which training rows the model sees, which is the variable the experiment manipulates. Each arm answers the specific question the previous one leaves open, which is why they are presented in the fixed order a–f. Two size-matched (c) designs exist and are kept separate (SI-C.1): a **nested 5 × 5 draw** (`diversity_control.py`, `outputs/composition_matched_control.csv`), whose statistic is the seed-level mean of that seed's five inner draws with the SD taken across the five outer seed means, and a **seed-42 repeated-draw** (`confound_diagnosis.py --size-match`, `outputs/isoreticular_control_sizematched.csv`); they are different replication designs, not two precisions of one estimate, and are never pooled into one error bar. Arm (d) reproduces (b)'s composition/topology profile, which confines it to (b)'s own row set, so it is an internal-consistency check rather than an independent mechanism contrast. The primary metric is MAE; RMSE and the deployable training-mean baseline are reported alongside.

Arm map, in one place. The seven arms with the row counts of the 50,000-sample protocol, so that no comparison is read across designs. **(a)** all training rows, 42,381 on the nbo side — the reference pool. **(b)** the partner topology removed, 10,656 — the total removal effect, read against (a). **(c)** a size-matched random draw of 10,656 rows from (a) — asks whether row count alone reproduces (b); it is read against (a) and (b), and its nested five-by-five and seed-42 repeated-draw variants are two replication designs of the same control, never pooled into one error bar. **(d)** a profile-matched draw confined to (b)'s own rows — an internal-consistency check on (b), not an independent arm. **(e)** only those partner rows whose composition recurs in the test set removed, about 12,342 rows on average — read against (a). **(f)** a size-matched draw from (e) — read against (e) and, through (e), against (b). **(g)** post hoc: a random removal of as many partner rows as (e) removes, drawn from (a) — the control for (e), run after a–f were fixed. Comparisons are read only between the arms listed in this map, within a seed-fixed test set.

2.5 Aggregation, replication and uncertainty

Grouped-split R^2 is reported under four rules: **pooled** (all fold predictions concatenated and scored once), **weighted** (per-fold R^2 weighted by test-set size), **macro** (unweighted mean of per-fold R^2) and **median**. All four are reported because no single rule is right for every deployment question — a whole-database deployment dominated by the large topologies is best described by pooled, a "typical per-topology" performance by macro or median — and they can differ in sign, so reporting one alone would hide the sign reversal. A fold whose test-set variance is zero makes R^2 undefined and is handled explicitly rather than silently dropped. In the original sample-redraw analyses, two distinct uncertainty objects are kept separate: the **standard deviation over seeds** describes how a quantity varies across independent redraws, while the **paired 95% interval** describes the confidence on a paired difference over the same seeds, and neither is rewritten as the other; a per-seed percentage spread is a third object and does not describe a paired effect. Percentages quoted for a paired change scale the absolute paired difference by the observed arm-(a) mean, with that denominator treated as fixed; they are not ratio intervals in which numerator and denominator are estimated jointly. Five seeds are five redraws within one database, not five independent databases or five independent topology populations, so these intervals describe seed-to-seed variation under that fixed database. Five-seed designs use seeds 42, 143, 244, 345, 446; five seeds are used rather than one because the screening and intervention quantities are seed-sensitive (the topology overlap standard deviation is 37% of its mean). Multiplicity is handled by declaration rather than adjustment. The paired t p-values are parametric summaries on five differences; the smallest attainable two-sided sign-flip p at $n = 5$ is 0.0625, so the intervals carry the inference and the p-values are not distribution-free evidence. The two prespecified primary paired tests — the nbo-side MAE contrast of §3.3 and the screening-recovery contrast of §3.4 — are single planned comparisons and are reported unadjusted; the remaining arm-level intervals, the ten-target panel, the ARC source tests and the per-topology correlations are descriptive and are not corrected for family-wise error.

2.6 Screening metrics and ARC sensitivity

Screening utility is measured on a support set shared within each seed: every sample carries an out-of-fold prediction from every strategy, so top-k overlap and regret are computed on identical rows and cannot be confounded by a change of denominator. These metrics pool out-of-fold predictions on the same candidate set; they describe retrospective validation-protocol performance rather than the shortlist of a single deployed model, because the contrast still includes differences in training-set composition and fold sizes even though the scoring rows are shared. Ties are resolved by a deterministic rule rather than by "take the first k": rows are sorted by score descending with a stable sort on the archived row order and the first k are taken, so a tie at the boundary is broken by row position — arbitrary but reproducible. On the true-uptake side no tie straddles the top-1%, top-5% or top-10% threshold in any of the five 25,000-row seeds (checked). The predicted scores were audited for the two strategies whose row-level predictions are archived (random and topology; five seeds, two row policies, 60 threshold checks): no tie straddles a cut, and replacing the archived-row-order rule with a lowest-source-row-ID rule changes no selected member (SI-F.1). The other three strategies are archived only as summary metrics, so this audit is partial. The **oracle** is the true top-k of the same samples, i.e. the materials a budget would select if the true uptake were known. k = 1% (250 materials at 25,000 samples) is the headline budget, chosen as a prespecified illustrative screening size; k = 5% and 10% are in the Supporting Information (SI-F). For ARC-MOF, two distinct controls are used and are not interchangeable: **within each source, topology labels are permuted across records while composition keys and topology-label counts are held fixed** (1,000 permutations), testing whether the observed composition–topology association exceeds that permutation reference; separately, a **common-rate binomial parametric bootstrap** (20,000 draws) assesses between-source heterogeneity. The two controls address different null hypotheses and neither identifies a database-construction mechanism (SI-H).

2.7 Follow-up designs and replication units

Six follow-ups (SI-I.1–SI-I.6; Tables S19–S24) examine, in this order, (1) representation, (2) constituent novelty, (3) CoRE structural feasibility, (4) fold-level support diagnostics, (5) matched partner replacement and (6) fixed-pool screening. The ARC-MOF audit is not one of the six: it is the second-database support audit reported in §3.5 (SI-H), because it adds no new performance or representation experiment. An additional section (SI-I.7) reports a prespecified feature-subsampling sensitivity for the two representation contrasts (Table S25) and is likewise counted outside the six. Representation and novelty analyses use shared samples and folds within each comparison: five 25,000-row redraws and one 100,000-row size check from the same database, and these are the same five draws used by the screening experiment, so the two are not independent replications of one another. G denotes the eight numerical descriptors; B adds metal-node and functional-group identities; I adds role-specific linker identities to B; T replaces those linker identities with mapped parent-linker Morgan fingerprints on a restricted common population. In the representation and novelty comparisons, RF and ExtraTrees use 120 trees and fold-fitted encoders. CoRE is a 40-structure technical pilot, with a 10-structure topology pilot. The risk analysis reuses folds from the original LOTO evaluation and therefore has eleven topology groups, not 66 independent groups. Partner replacement fixes one sampled test population and background, approximately matches numerical descriptors and varies shared-composition rows at constant training size and partner fraction. Its five repeats vary replacement order and forest randomness. Fixed-pool screening uses the cleaned replacement endpoints with G/B/I and both ensembles; five seeds vary learner randomness on fixed candidates and training memberships. These repetition units differ from the five independent sample redraws of §3.4. Full methods, residual imbalance and verification limits are stated in SI-I.

3. Results

3.1 Chemical support removed by topology hold-out

On the full hMOF database (324,426 structures), **74.7% of coarse compositions occur in a single topology**, and those compositions account for **61.3% of structures**. The two numbers answer different questions and must not be interchanged. The first weights compositions equally: of the 1,622 coarse compositions (metal centre plus the two linkers), 1,212 occur in exactly one topology, so nearly three-quarters of the defined composition-key space is private to one topology. The second weights structures: those 1,212 single-topology compositions account for 61.3% of the 324,426 structures, so a majority of the database loses its same-key training counterpart when its topology is held out. They differ because single-topology compositions are on average less populous than multi-topology ones, and quoting only 74.7% invites the false reading that 74.7% of structures are unseen. Both units are reported throughout because the distinction is the difference between "most chemistries" and "most structures".

The loss is strongly heterogeneous across topologies (Figure 1). Seven of the eleven topologies have no other topology sharing any of their composite chemistry keys, so holding one out leaves every structure of that topology unseen — a **complete** chemistry hold-out, in the sense that the composite key is wholly absent from the remaining training set. The seven are sra, etb, acs, bcu, pts, fof and the, and together they hold only **21.3%** of structures (23,962 + 19,682 + 13,655 + 8,795 + 1,522 + 1,256 + 352 = 69,224 of 324,426); "7 of 11 are exact hold-outs" is therefore not "most structures are exact hold-outs", and the two facts must be stated together to avoid overstating how much of the database is affected. The remaining four topologies are covered to differing degrees: pcu leaves 60.1% of its structures unseen, nbo 12.3% and tbo 89.0%, while rht (14 structures, sharing two compositions with tbo) is fully covered. The complete eleven-row decomposition and the metal/linker/conjunction breakdown are in SI-B; the metal/linker/conjunction split shows that neither the metal centre nor the linker pair alone reproduces the conjunction's exclusivity, but the headline remains the composite key because that is what the topology hold-out actually withholds. The point the main text keeps is that "completely unseen" holds only for the composite chemistry key, not for each building block separately — a topology can have no composite-key counterpart while its metal centre or a linker still recurs in training.

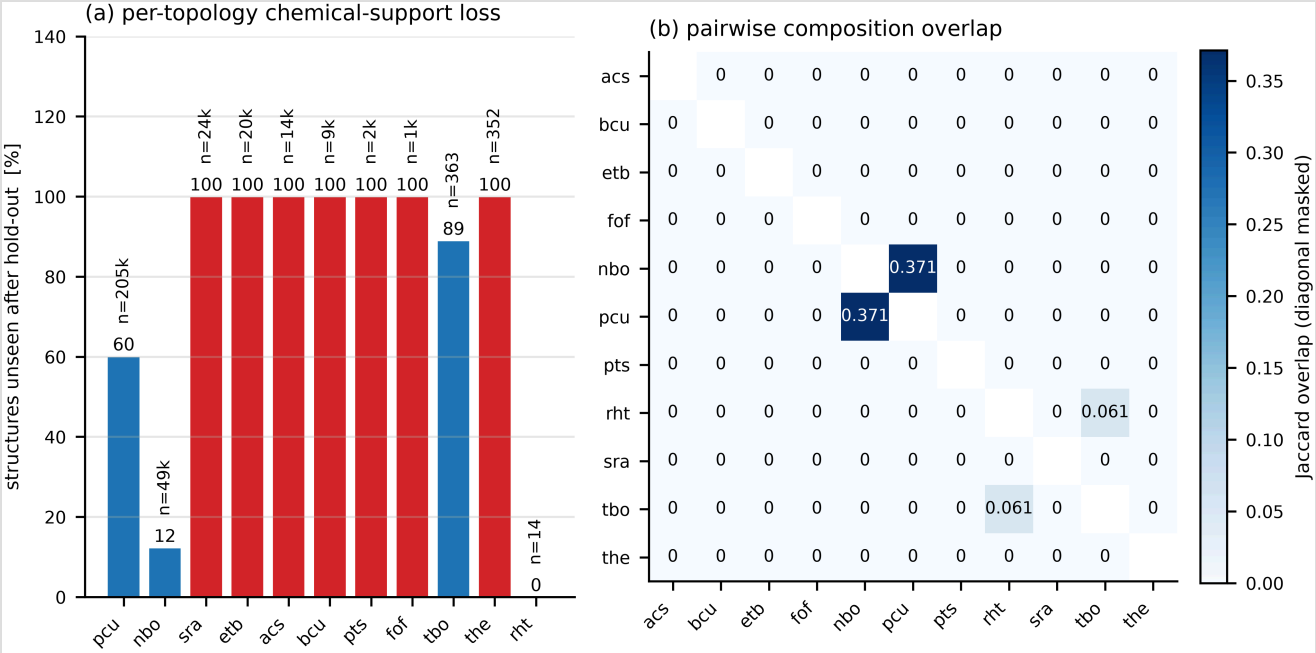


Figure 1. Chemical support removed by topology hold-out on hMOF. **(a)** Per-topology share of structures whose coarse composition key has no training counterpart after that topology is held out; each bar is labelled with that percentage and with the topology's structure count in thousands (n = 205k … 14), and seven of eleven topologies sit at 100% but hold only 21.3% of structures, while pcu (60.1%), nbo (12.3%) and tbo (89.0%) are partially covered and rht is fully covered. **(b)** Pairwise Jaccard overlap of the coarse composition sets; the diagonal is masked (self-overlap is not zero), and only pcu ↔ nbo (0.371) and rht ↔ tbo (0.061) are non-zero, leaving one usable contrast. *Source data:* `outputs/chemistry_seen_by_topology.csv`, `outputs/topology_pair_overlap.csv`.

Conditioning on composition removes **82.0%** of the entropy in the topology label (H(T|C)/H(T) = 0.180, plug-in estimate, structure-weighted; the first-order Miller–Madow correction to H(T|C) is 0.000912 bit, i.e. 0.3% of the 0.320-bit estimate, so the headline is unaffected at this granularity), consistent with strong statistical coupling between the specified composition keys and topology labels. The relation is asymmetric and both directions are reported: the reverse ratio H(C|T)/H(C) = 0.860 says topology leaves 86% of composition uncertainty, so 18.0% is the small direction of an asymmetric relation, not a symmetric measure of mutual determinism. The weighting matters and is stated once: the ratio is structure-weighted throughout, so the composition-weighted value (11.4%) differs from the headline because its numerator weights compositions equally while its denominator is the structure-weighted H(T), so it is a descriptive contrast with 18.0% and not an alternative estimate of the same quantity; quoting 18.0% alone would invite the false reading that the two variables mutually determine each other. The fine-granularity figure moves in the same direction (93.2% single-topology; 84.6% structures unseen), but for a different reason — requiring an exact functional-group match creates many singleton compositions that are single-topology by construction rather than by observation. Restricting to compositions that occur at least twice lowers the fine single-topology fraction to **79.1%** (66,172 non-singleton compositions), whereas the coarse headline is barely affected (only 21 of 1,622 coarse compositions are singletons; 74.4% among non-singletons against 74.7% overall). The fine-granularity robustness check is therefore doing exactly one job — confirming that the coupling's direction is not an artefact of the coarse key — while leaving its magnitude to the coarse headline, and the two values are not interchangeable. Fine-grained values are reported only as robustness checks with these caveats, not as headline evidence (SI-B).

A weaker version of the "by construction" objection applies to the headline itself, and is stated here rather than left to the reader. hMOF is enumerated by recombining finite building-block sets on a library of topological templates (ref. 1), so some association between a coarse composition key and a topology is expected from the recipe, and 74.7% should not be read as an independent discovery about MOF chemistry. What the audit supplies is a deployable instrument: for a reporter's own library and key, it measures the support a specified split removes, from a table the reporter already holds. **The instrument reports a quantity, not its cause: on an enumerated library such as hMOF, a high single-topology fraction is partly expected from the building-block recombination recipe, so the output must not be read as evidence about MOF chemistry, and a reporter should state whether their own library is enumerated before quoting the fraction as a finding.** Separating how much of the coupling is fixed by the enumeration recipe from how much reflects chemistry beyond it is not attempted here: the second audit (§3.5) uses different keys on a partly dependent library, so it cannot supply that separation.

For a practitioner, the practical reading is already fixed by these counts. A topology hold-out on this database is, in seven of eleven cases, an exact composite-key hold-out, so a score from such a fold is not "transfer to an unseen topology" but "transfer to an unseen topology-and-chemistry combination". In the remaining four cases it is a partial chemistry hold-out whose severity the per-topology unseen fraction quantifies. This remains an evaluation of held-out topology groups, with an additional composition-key shift; the counts alone do not establish its difficulty or its expected error.

The pairwise-overlap profile explains why the controlled experiment is possible at all. Of the 55 topology pairs, only pcu ↔ nbo (408 shared compositions) and rht ↔ tbo (2) share any composition; every other pair has Jaccard = 0.000. The rht ↔ tbo pair is unusable, so a single pair — pcu ↔ nbo — carries the entire controlled experiment, and the training-intervention contrasts are confined to that one pair. The 408 shared compositions are not a bare label but a usable quantity of data: in the 50,000-sample protocol, deleting them in arm (e) removes 12,342 training rows on the nbo test side and 6,689 on the pcu test side, enough to move the fit detectably, which is what separates this pair from the two-composition rht ↔ tbo pair by two orders of magnitude. Two further facts bound how the headline is read. First, the fine-definition single-topology statistic is sample-size dependent — 98.9% at n = 25,000 against 93.2% at the full database, for the reason given in §2.1 — so the full-database number is the headline and every value carries its n. Second, a negative result closes the section: the per-topology unseen-chemistry rate does not detectably order the per-topology hold-out cost (Pearson $r = +0.10$, $p = 0.78$; Spearman $\rho = +0.28$, $p = 0.44$ across the ten topologies with $n_{test} \geq 100$; SI-B). Support loss can therefore be reported as a structural fact, but it is not converted into a predicted error magnitude: with ten topological units this is a low-power null, not evidence that no relation exists, and the direction of any error ordering is not supported. This negative result is what forces the paper to earn its performance claims empirically in §3.2 and §3.3 rather than inferring them from the support counts.

3.2 Protocol contrasts distinguish support loss from performance loss

On 100,000 samples with the shared 426-feature set, the five validation schemes give pooled R^2 of 0.7529 (random), 0.6738 (chemistry key), 0.6606 (linker pair), 0.5270 (metal) and 0.3786 (topology, 5-fold GroupKFold) (Table 1). Each scheme is evaluated in one five-fold cross-validation run, fitting a 120-tree random forest separately in each fold on the same 100,000-row sample at seed 42, without seed averaging or hyperparameter tuning; the 415-feature variant removes the topology one-hot block from both arms (SI-A.2b, SI-D). The observed order is chemistry key (0.674) > linker pair (0.661) > metal (0.527) > topology (0.379); this is a list of measured values, not a monotonic "coarsening" law, because the topology grouping withholds a structural family as well as its chemistry and is not simply a coarser member of the same chain. The chemistry-key hold-out therefore removes far less of the R^2 than the topology hold-out: holding out the composite chemistry key retains 0.674 against topology's 0.379. The two are not on identical footing as they stand — the chemistry arm keeps the 11 topology one-hot columns that the topology arm loses by construction, and the two protocols differ in fold structure and training size (the topology arm groups into 11 topologies over 5 folds, training on 36,553 to 92,965 records, while the chemistry arm groups into 1,590 composition keys over 5 folds, training on 80,000 each) — so the comparison was repeated with the topology block removed from **both** arms. One asymmetry survives that control: the fold holding out pcu supplies 63,447 of the 100,000 evaluation rows and is trained on 36,553, so the pooled topology R^2 is dominated by its smallest training set, and the contrast is read as an ordering of constructions rather than as a size-matched effect.

The chemistry-key decrease is **21.1%** of the topology-grouped decrease with 426 features and **21.5%** with matched 415-feature sets (matched decrease 0.3603; Figure 2a). The 415-feature check exists to rule out one specific objection — that the chemistry arm is information-advantaged because it keeps the topology one-hot block that the topology arm loses by construction. Removing that block from both arms moves the chemistry arm by only −0.0065 R^2, so the ordering is not an artefact of the unmatched feature count. These are **descriptive ratios between validation protocols**, not causal shares of the penalty: the two protocols differ in fold structure, training-set size, chemical coverage and group structure, and matching 415 features does not match any of those. The complement of the ratio is deliberately not named as a mechanism — neither "chemistry" nor "geometry-to-uptake mapping" is claimed to account for the remaining ~79%, because the design does not identify it. Under this representation and these fold constructions, the two protocol decreases differ substantially, and the ratio is read as an ordinal contrast between constructions rather than as a population-level share. This contrast does not isolate a component beyond chemistry or determine how much of either decrease is caused by chemical novelty; the representation checks in §3.6 further qualify its interpretation.

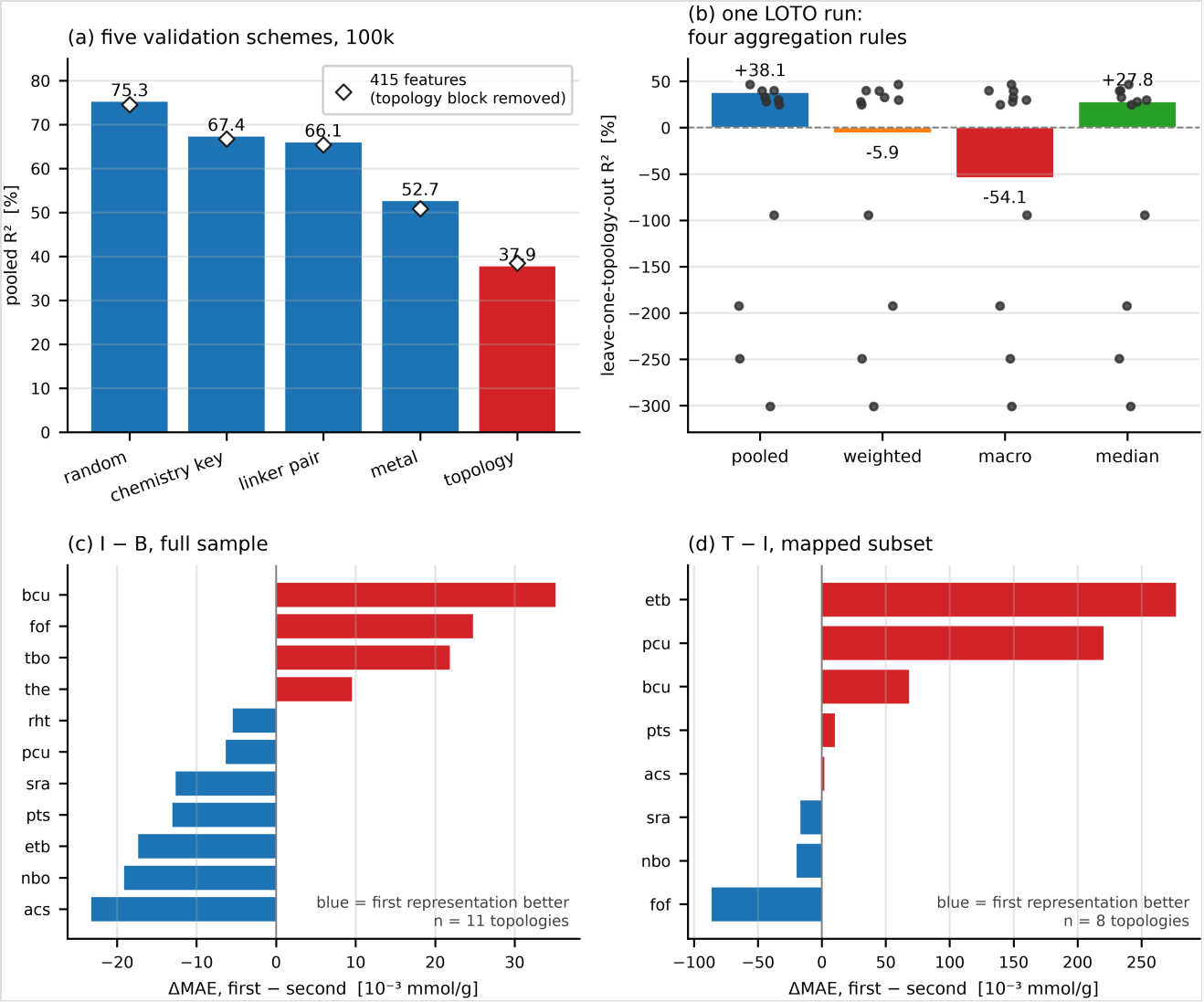


Figure 2. How the support change relates to measured performance on the 100,000-sample run. **(a)** Pooled R² of the five validation schemes (bars, 426 features) with the matched 415-feature check overlaid as diamonds (topology block removed from both arms); the chemistry-key arm reaches 0.674 against 0.379 for the topology arm, and the 415-feature check moves neither ordering nor the 21.1%/21.5% ratio. **(b)** One leave-one-topology-out run scored under four aggregation rules (bars) with the eleven per-fold R² values overlaid as dots; the sign of the aggregate depends on the rule, and the per-fold spread is why no single aggregate is reported alone. **(c)** Per-topology paired MAE change for I − B on the full sample (five-redraw mean, uptake, RF), and **(d)** for T − I on the mapped subset (eight topologies); blue marks a lower MAE for the first representation, red a higher one. The pooled improvement in (c) and the pooled degradation in (d) each cover topologies that move the other way. *Source data:* `outputs/leakage_results_final.csv`, `outputs/leakage_results_chemistry.csv`, `outputs/leakage_results_notopo100k.csv`, `outputs/loto_aggregates_100k.csv`, `outputs/loto_per_topology_metrics_100k.csv`, `outputs/finish_20261023/per_topology_paired_deltas.csv`.

The additional model-family checks also show lower topology-grouped than random-split R^2. Each family contributes one cross-validation run per protocol on the same 100,000 rows, the same 426 features and seed 42, with a separate fit in each fold and no seed averaging or outer hyperparameter search: a 120-tree random forest, a histogram gradient-boosting regressor with default settings, and a ridge regression that selects alpha internally from 0.1, 1, 10 and 100 using `RidgeCV` after standardisation (SI-A.2b, SI-D.2). The two model families that reach comparable random-split accuracy agree on the topology penalty to within 0.0001 R^2 (0.3743 for random forest against 0.3742 for gradient boosting), and ridge shows a smaller absolute gap (0.4401 → 0.2603, a penalty of 0.180) that is largely a level difference — its random-split R^2 is 0.440, against 0.753 for the random forest. Normalising each family by its own random-split R^2 — as a loss fraction, (random − topology)/random — gives 49.7% for the random forest, 51.1% for gradient boosting and 40.9% for ridge, i.e. retained fractions of 50.3%, 48.9% and 59.1%, so the normalisation does not bring the three together and no claim of model equivalence is made; these are single cross-validation runs with the model-selection procedures stated above (SI-D). The penalty is also target-dependent — the same design on log-selectivity gives a smaller gap (0.8357 → 0.7211 against 0.7529 → 0.3786 for uptake) — so the magnitude quoted here is a property of the 0.15 bar, 298 K uptake target, not a universal constant of the protocol, and no single magnitude should be quoted without its target.

Leave-one-topology-out evaluation (a different protocol from the five-fold GroupKFold above) gives pooled R^2 = +0.3814 against weighted −0.0589, macro −0.5414 and median +0.2782 (Table 1, Figure 2b). This is one run at seed 42 of a 120-tree random forest, with one model per held-out topology (eleven models); the five-fold GroupKFold value 0.3786 in the same panel is a different fold construction trained on a different pool, so the two are not two precisions of one estimate. The two protocols are not interchangeable: the 5-fold GroupKFold trains five models on the folds of the 100,000-sample design, whereas LOTO trains one model per held-out topology (eleven models) on all other topologies and scores that topology alone, so their training pools and test folds are different objects. The sign of the aggregate depends on how folds are weighted: pooled scores every fold's prediction once against a single overall mean, while macro averages per-fold R^2 each normalised by its own variance, and a zero-variance fold makes R^2 undefined. Two different fold counts are involved and must not be merged: on the 100,000-sample single LOTO run there are eleven scoring folds, while the 25,000-sample multi-seed runs of SI-D use 10–11 valid folds per seed because a fold with zero target variance can drop out. Dropping the 5-structure rht fold moves the 100k macro from −0.5414 to −0.2948. The headline aggregation is chosen to match the intended deployment population, and the per-fold values are reported alongside; the positive pooled value is not produced by the pcu fold alone, as recomputing pooled while dropping only the pcu scoring fold (model and training folds unchanged) gives +0.4629. The per-fold values and the five-seed stability are in SI-D.

3.3 Fixed-test-set interventions reveal asymmetric partner effects

The pcu ↔ nbo contrast fixes the test set within each seed and changes only the training pool (Methods 2.4). MAE is the primary metric here because it is the deployment-relevant quantity — the average error in mmol/g that a screening model would carry — and it stays defined for the small single-topology test sets where R^2 is volatile; RMSE and the training-mean baseline are reported alongside, and the metric does not change the conclusion. The six arms are designed to answer one question in sequence — *is the degradation caused by how many training rows are removed, or by which ones?* Arm (a) is the full training pool (42,381 rows for nbo on average). Arm (b) removes the partner topology, leaving 10,656 rows on average. Arm (c) keeps the row count of (b) but draws the rows at random from (a), so it matches the size but not the identity of the removal. Arm (d) keeps the composition/topology profile of (b), which as shown below confines it to (b)'s own rows. Arm (e) removes only the partner rows whose composition recurs in the test set, and arm (f) is its size-matched draw. The comparisons are read as follows: (a)→(b) is the total effect of removing the partner; (a)→(c) asks whether random thinning to the same training size reproduces that effect; (b)→(c) asks which rows are retained at a fixed size; (e)→(f) asks the same size question inside arm (e)'s pool; and (b)→(f) is the size-matched version of the (e)→(b) comparison. No comparison isolates a single chemical mechanism.

On the nbo test set, removing the partner pcu raises MAE from 0.1065 ± 0.0032 to 0.1811 ± 0.0136 mmol/g — a paired difference of **+0.0746 mmol/g** (paired t-interval +0.0591 to +0.0900) — which is **+70.0%** when scaled by the observed arm-(a) mean with that denominator treated as fixed, and **+55.5% to +84.5%** when the same interval bounds are divided by 0.1065 (paired $t(4) = 13.4$, $p = 1.8 \times 10^{-4}$) — and RMSE rises by +0.1053 mmol/g (+77.9%, same scaling) (Figure 3, Table 2). The interval is a t-interval on the five absolute paired differences, not a ratio interval and not a per-seed percentage spread; because the numerator and the denominator are not estimated jointly it does **not** carry nominal 95% coverage, so the percentage bounds (+55.5% to +84.5%) are a descriptive spread — that same interval divided by the fixed denominator 0.1065 — and not a confidence bound on the percentage itself. Its lower bound excludes zero, so the degradation is not within seed noise, and the interval is reported rather than a single point so the reader can see the effect's stability. A size-matched arm (c) that draws the same number of training rows at random from the full pool leaves MAE essentially unchanged (c: 0.1071 ± 0.0048, five outer seeds × five size-matched draws, averaged within seed before the seed-level SD; the seed-42 repeated-draw auxiliary c gives 0.1030 ± 0.0031, a distinct replication design kept in SI-C). The size-matched arm keeps 1,484 of the 1,532 compositions that arm (a) carries, so it preserves the chemical diversity while dropping the row count; its MAE is close to arm (a)'s (0.1071 against 0.1065), not to arm (b)'s (0.1811), so this random-thinning control does not reproduce the degradation. What differs between (b) and (c) is the identity and distribution of the retained rows at a fixed training size. The random-thinning result indicates that row count alone does not reproduce this contrast; it does not exclude interactions between training size and the distribution of retained rows. Both arms still beat their respective training-mean baselines on the nbo test set, so the +70.0% describes a larger absolute error on the same test values (baseline errors and seed-level variability in SI-C). Neither arm beats the test-mean predictor on this fold: R^2 is −0.875 for arm (a) and −4.383 for arm (b), against +0.268 and +0.262 on the pcu side (SI-C Table S4), so the percentage describes error growth on a fold that no fit here predicts well.

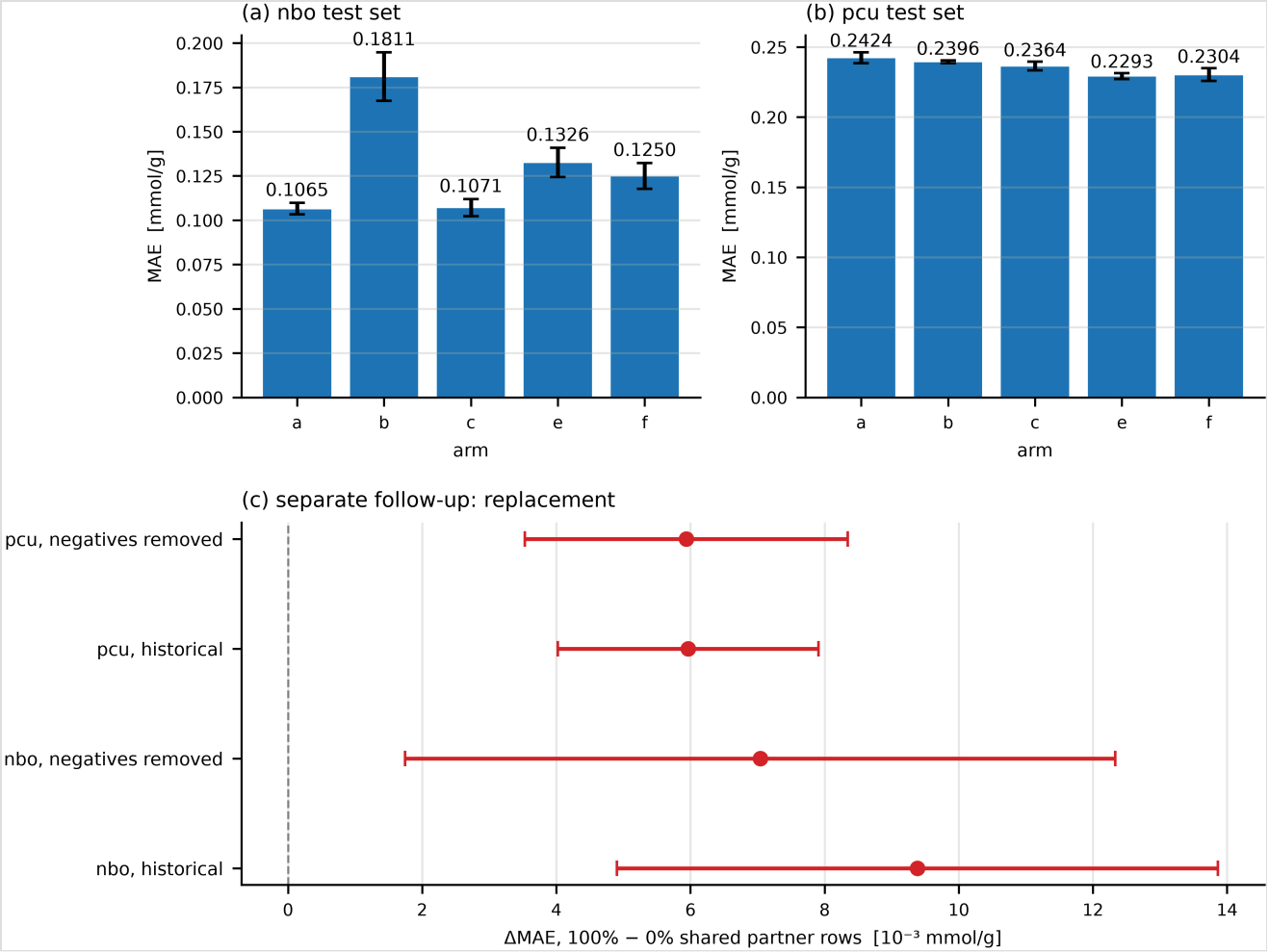


Figure 3. Fixed-test-set interventions on the pcu ↔ nbo pair (five outer seeds; test set fixed within each seed). **(a)** nbo-side and **(b)** pcu-side MAE per arm; bars are five-seed means, error bars ±1 SD across the five outer seed means (arm (c) is the nested 5 × 5 draw, its SD taken across seed means). Arm (d) is not plotted because it is confined to arm (b)'s own row set and therefore reproduces (b) by construction (SI-C.2); the post hoc arm (g) is reported in SI-C.4 and also not plotted. On the nbo side, removing the partner (b) raises MAE by +70.0% while the size-matched draw (c) does not, and the shared-composition removal (e) and its size-matched draw (f) fall in between; on the pcu side arm (e) improves MAE relative to (a), so the effect is direction-dependent. **(c)** A separate follow-up design, not an arm of the a–g contrast: matched partner replacement at the two endpoints, on its own test population and with its own five learner seeds (SI-I.5). It shows the paired MAE change from 0% to 100% shared partner rows, with 95% intervals, for the historical and negative-descriptor-removed row policies; both directions move the same way and both intervals exclude zero, but residual composition differences prevent a causal reading. *Source data:* `outputs/isoreticular_control_seeds.csv`, `outputs/composition_matched_control.csv`, `outputs/arm_e_partner_coverage.csv`, `outputs/followup_20261023/experiment5/paired_endpoints.csv`, `outputs/followup_20261023/experiment5/cleaned_endpoint_summary.csv`.

The same intervention is not symmetric. On the pcu test set, removing the partner nbo leaves MAE essentially unchanged (−0.0028 mmol/g, paired t-interval −0.0080 to +0.0024, i.e. −1.2% of the arm-(a) mean) and RMSE essentially unchanged too (0.4030 ± 0.0021 to 0.4024 ± 0.0030, −0.1%), so the two sides of the pair behave differently from the outset: nbo's MAE degrades heavily when its pcu partner is removed, while the pcu-side change is small and its paired interval includes zero — the pcu side is far less sensitive to the partner, not independent of it. Both arms also beat their training-mean baselines on the pcu side; that comparison bounds them against a constant predictor, but beating a weak baseline does not exclude the model approaching its own error floor, so the near-zero change is read as no detected difference rather than as evidence of equivalence. This asymmetry foreshadows the direction dependence of the later arms. In this retrospective test, the increase is from approximately 0.107 to 0.181 mmol/g; it is not an estimate for a new deployment population. That increase could matter for screening, but the ranking effect of this intervention was not measured; §3.4 measures ranking under a separate 25,000-sample protocol contrast.

Two further arms qualify the result. Arm (e), which removes only those partner rows whose composition also occurs in the test set, raises nbo-side MAE to 0.1326 ± 0.0082 — more than the full pool but far less than removing the whole partner (0.1811 ± 0.0136) — while a size-matched draw (f) gives 0.1250 ± 0.0073, essentially unchanged from (e) and still far better than (b), showing that a random draw from the (e) pool remains more accurate than (b) at the matched size. The 34.9% / 65.1% split of the (a)→(b) degradation — computed from the archived five-seed means as (e)−(a) = 0.02605 over (b)−(a) = 0.07459 mmol/g, i.e. 34.92% — is an **ordering of magnitudes inside one contrast**, not a decomposition into two independent causal factors, and the two steps remove different coverage rather than a single shared quantity: arm (e) deletes the 12,342 partner rows that carry the 388 compositions shared with the test set (387 at the seed-42 baseline), leaving 1,144 of 1,532 compositions and 30,039 of 42,381 rows, while the further (e)→(b) step deletes the remaining partner rows and loses a different 648 compositions (the remaining 65.1% of the degradation). That step also drops 19,383 training rows (30,039 → 10,656), so the 65.1% residual mixes composition loss with a size reduction; the size-matched arm (c) puts the size contribution near zero on this pair (0.1071 ± 0.0048 at 10,656 rows against 0.1065 ± 0.0032 at 42,381), which is why the residual is not read as a composition effect. Recomputing the split from the four-decimal Table 2 entries gives 35.0%; the 0.1 pp difference from the printed 34.9% is therefore rounding, not a discrepancy, and the archived five-seed means are the basis for the printed values. Crucially, the effect is **direction-dependent**: on the pcu test set arm (e) improves MAE to 0.2293 against arm (a)'s 0.2424, and the pcu-side size-matched draw (f) gives 0.2304 ± 0.0045, still better than arm (b)'s 0.2396, so random size matching does not reproduce the observed directional contrast. The paired pcu (e)−(a) difference is −0.0131 mmol/g (95% CI −0.0167 to −0.0096) and the (f)−(b) difference is −0.0092 mmol/g (−0.0144 to −0.0040); both intervals exclude zero within this five-redraw design. Post hoc, arm (g) removes the same number of partner rows at random, without conditioning on shared composition; it lies between (a) and (e) on both sides — pcu 0.2341 against a 0.2424 and e 0.2293; nbo 0.1189 against a 0.1065 and e 0.1326 — but the paired (e)−(g) interval includes zero on pcu (−0.0101 to +0.0004 mmol/g) while remaining positive on nbo (+0.0090 to +0.0183; SI-C.4). Thus the point ordering does not establish a shared-composition-specific effect in both directions, and none of these arms identifies different physical adsorption mappings. Arm (d), the composition-profile-matched draw, reproduces (b)'s composition/topology profile, which confines it to (b)'s own row set (five-seed means: 42,381 − 31,725 = 10,656; seed 42 gives 42,329 − 31,682 = 10,647, SI-C.2); its aggregated MAE is the same as (b)'s by construction (nbo 0.1811 ± 0.0136; pcu 0.2396 ± 0.0010), so it is a construction-consistency check, not a second independent mechanism contrast (SI-C).

The original arms show a local nbo-side partner-removal effect, a random-thinning contrast and different responses across the two directions. The descriptive 34.9% / 65.1% steps do not isolate coverage and diversity. A stricter follow-up fixes training size, partner fraction and background while replacing approximately descriptor-matched non-shared partner rows with shared-composition rows. Relative to the 0%-shared endpoint, the 100%-shared endpoint increases MAE by 0.0094 mmol/g on nbo and 0.0060 on pcu; both paired intervals exclude zero, and the signs persist after rows with negative numerical descriptors are removed (Table S23). This differs from a simple expectation that more shared composition should improve accuracy. Functional-group and metal-node distributions remain imbalanced, and the repeats use the same test population; the result is a constrained within-pair contrast, not a pure chemical causal effect. Full original per-seed rows are in SI-C and the replacement design is in SI-I.5.

Two boundaries belong to this result, not only to the general discussion. The first is the pair: of the 55 topology pairs only pcu ↔ nbo carries usable shared-composition support, and the second candidate (rht ↔ tbo, two shared compositions) cannot test anything, so the asymmetry is a property of this pair's two directions rather than a relation shown to generalise across pairs. The second is the library and the target: every arm, the replacement included, redraws one enumerated library at one target (0.15 bar, 298 K uptake), so +70.0% is a within-library model response and carries no information about library-to-library or experimental-material variation — the five redraws vary the sample, not the database. Neither boundary is removable with the data used here, which is why the later follow-ups are read as limits on interpretation rather than as replication.

3.4 Screening utility on a shared support set

This section confirms and extends the screening result of Asgari et al. (ref. 16). The direction — a grouped split collapsing top-k recovery relative to random splitting — is theirs, reported there at a top-5% budget; what is added here is a top-1% budget, the shared-support-set construction and a paired test across five redraws.

Across five independently redrawn samples (25,000 rows; seeds 42, 143, 244, 345, 446), random-split validation recovered **50.4 ± 2.5%** of the true top 1% and topology-grouped validation **19.3 ± 7.1%** (mean ± SD), a paired difference of 31.1 percentage points (SD 5.3, 95% paired t-interval 24.6 to 37.6 percentage points; $t(4) = 13.2$, $p = 1.9 \times 10^{-4}$) (Figure 4). The paired test is on the five within-seed differences, so it inherits the seed-to-seed scope set out in §2.5 and is not evidence about other databases or other topology populations. These rankings pool out-of-fold predictions from several fitted models on rows shared across strategies. Predictions from different folds are not standardised before pooling, so the contrast includes between-fold calibration differences as well as support loss. The 1% budget is an illustrative retrospective evaluation budget, not an empirically established synthesis capacity. Intermediate recoveries were chemistry key 44.0 ± 2.8%, linker pair 41.1 ± 4.1% and metal-node group 29.2 ± 4.8%. This ordering resembles the R^2 comparison, but differing folds and training populations prevent attributing the recovery gap to a single chemical or topological factor. Random ranked best and topology worst in all five redraws; the complete ordering held in four. These are within-database protocol contrasts and do not estimate a deployed model's prospective hit rate.

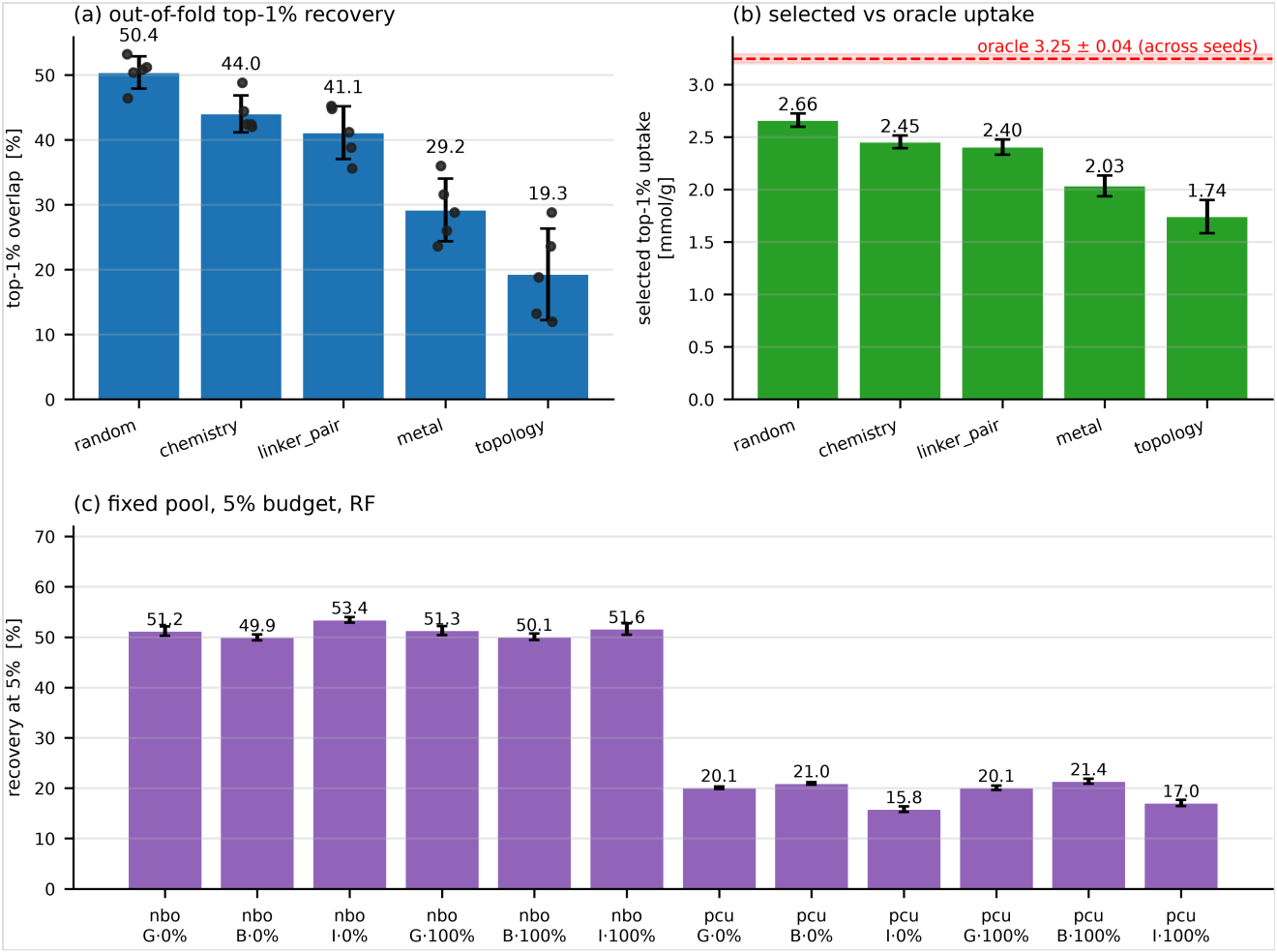


Figure 4. What the protocols mean for screening. **(a, b)** Retrospective out-of-fold screening on five 25,000-sample redraws; each ranking pools predictions from several models. **(a)** Top-1% overlap for the five strategies; bars are five-seed means, error bars ±1 SD, and the filled dots are the per-seed values on the shared within-seed support set — random best and topology worst in all five seeds, the complete ordering holding in four. **(b)** Mean true uptake of the selected top-k in the same strategy order; the dashed line is the mean oracle uptake (≈3.25 mmol/g) from the same seeds. **(c)** Fixed candidate pool at the illustrative 5% budget (RF), recovery for G/B/I at 0% and 100% shared partner rows in both directions; the same representation change helps on one side and hurts on the other, so a single-model shortlist is not implied by (a, b). *Source data:* `outputs/screening_utility_seed{42,143,244,345,446}.csv`, `outputs/followup_20261023/experiment6/screening_summary.csv`.

Materials selected under topology-grouped validation averaged 1.742 ± 0.158 mmol/g, **46.3% below** the mean oracle uptake (one minus the ratio of mean selected to mean oracle uptake; the per-seed mean of ratios rounds to the same 46.3% but is a distinct statistic). Overlap and regret are complementary: overlap counts how many of the true top-k were found, regret is the oracle uptake minus the selected uptake. At the seed-42 baseline the topology arm selects 1.587 mmol/g against an oracle 3.319 (regret 1.732), while random selects 2.706 (regret 0.613). Widening the budget to k = 5% and 10% attenuates but does not erase the gap (full table in SI-F).

Topology overlap ranges from 0.120 to 0.288 across redraws (SD 0.071), whereas the random-split SD is 0.025. The 31.1-percentage-point gap corresponds descriptively to 61.7% fewer hits in the topology OOF ranking. The random-split result is not an attainable benchmark for deployment to a topology with no in-topology training data.

The fixed-candidate-pool follow-up evaluates each fitted model on one frozen candidate population per direction (Table S24). At a 5% budget with RF and no shared-composition partner rows, changing G to I raises nbo recovery from 51.20% to 53.44% but lowers pcu recovery from 20.08% to 15.81%. With I, replacing the non-shared endpoint by the shared endpoint changes recovery from 53.44% to 51.61% on nbo and from 15.81% to 17.05% on pcu. Thus representation and partner changes do not produce a uniform screening gain, and prediction error and top-k recovery should be assessed separately. These are retrospective results on two fixed pools, with five learner seeds, rather than prospective synthesis outcomes or replication across candidate populations. The tie audit behind the rankings is partial: row-level predictions are archived for the random and topology strategies only, so the tie behaviour of the other three is known from their summary metrics rather than checked row by row (SI-F.1).

3.5 A bounded support audit in ARC-MOF

The same support audit on ARC-MOF (264,225 structures, of which 256,351 carry both metal and linker cluster keys — the other 7,874 lack them and are excluded; topology and cluster metadata only; Figure S8a–c and Table 3) gives a coarse single-topology fraction of 99.3%, which remains **97.3%** when restricted to the 49,869 non-singleton compositions, and a fine non-singleton rate of **98.2%** (26,380 non-singletons) (Table 3). Restricting to non-singletons matters because a singleton composition is single-topology by construction; the high value therefore does not reduce to a sparse-count artefact, and the non-singleton figure is the more resistant claim. The association persists after singleton removal and under the finer key; these checks concern sparsity and key definition, not independence from hMOF. The fine figure runs higher than the coarse one for the same reason it does in hMOF — the functional-group key fragments the composition space into more, smaller cells, each of which is more likely to be single-topology — and it is reported next to the coarse value precisely so that the reader does not mistake the finer key for a cleaner separation. The same entropy-correction audit that demotes hMOF's fine figure to a robustness check is applied here: the conditional-entropy correction is $(K_{TC} - K_C)/(2 N \ln 2)$, with occupied marginal/joint cell counts of 180,989/184,978 (coarse) and 177,971/178,834 (fine), giving corrections of 0.0112 bit (15.2%) and 0.0029 bit (19.8%) against $H(T|C) = 0.0739$ and 0.0147 bit, so both ARC-MOF values survive the correction. The two shares are computed before rounding — 15.188% and 19.795% — so 15.2% and 19.8% are correct; dividing the four-decimal entries (0.0029/0.0147) gives 19.7% and understates the second share by 0.1 pp. The audit consumes only topology and cluster metadata, so it is a support audit, not an external prediction test on adsorption properties ARC-MOF does not carry in the form this study's models consume.

ARC-MOF is **not independent of hMOF**: it describes itself as spanning public MOF databases and shares authors with hMOF, so a zero file-name intersection proves nothing about content overlap. The source-stratified audit is possible at all because ARC-MOF ships its own provenance as file-name prefixes (DB0 … DB15), a metadata affordance that enumerated libraries should provide but which is not guaranteed elsewhere. Removing one source sub-library at a time (the DB0 condition stated rather than assumed away) bounds the effect of any overlap. Deleting the largest sub-library DB0 (201,093 rows) moves the single-topology fraction to 98.57% (coarse) and 98.96% (fine); it is also by far the most expensive removal, discarding 79.0% of the analysed rows — 169,489 of 214,543 at the fine granularity — and leaving 45,054 rows (21.0%), so this is a stress test on about a fifth of the data, not a worst case over all of it. Removing any single *other* sub-library leaves the fraction between 99.22% and 99.46% at the coarse granularity and between 99.72% and 99.84% at the fine granularity, and none costs more than 12,496 rows (DB5, 5.8% of the analysed rows), against 169,489 for DB0. The bound holds only if the hMOF-derived rows are concentrated in DB0; that condition is stated here but **not demonstrated**: the only join available between the two archives is the file-name prefix, and a zero file-name intersection proves nothing about content overlap (as noted above), so if those rows instead spread across sources the bound loosens. Within each source, the observed single-topology fraction sits above its own topology-permutation null on a common support set, with excesses from +2.6 (DB15) to +37.3 (DB12) percentage points (for example, DB0 observes 99.7% against a null of 77.2%, DB12 observes 98.4% against 61.1%, DB15 observes 85.4% against 82.8%), and the degenerate single-topology source excluded. The excess shows a moderate-to-strong positive point estimate against the source's topology entropy (Pearson $r = +0.67$, $p = 0.10$), but there are only seven non-degenerate sources, so this is weak evidence either way; removing the single influential source DB12 flips the point estimate (Pearson $r = -0.08$, Spearman ρ -0.09), so the association is not established. The source-to-source spread itself is informative: at the common-support coarse granularity the exclusivity rate varies from 85.4% to 100.0% (86.1% to 100.0% at the fine granularity), so the audit reveals source-associated heterogeneity rather than a single number shared by all enumerated libraries, and that heterogeneity is why the per-source audit, not a single database-wide value, carries the ARC-MOF claim. The audit does not identify which construction choices caused that heterogeneity. As in §3.1, this is a report of a quantity, not of its cause: ARC-MOF is a mixed-source library rather than an enumerated one, but its per-source exclusivity spread (85.4% to 100.0% at the common-support coarse granularity) is source-associated rather than a single property of MOF chemistry — the construction choices behind that spread are not identified here — so a high single-topology fraction cannot be read as evidence about MOF chemistry, and the same caution transfers to any mixed-source library a reporter audits.

Two controls prevent the audit from being read as stronger than it is, and they address different null hypotheses. The leave-one-source-out range (98.6–99.8%) is a partial, not a complete, bound on cross-source overlap, for the DB0 condition stated above. The **within-source control** permutes topology labels across records within each source, holding composition keys and label counts fixed (1,000 permutations), and rejects "the observed association is what a within-source label shuffle would produce". Separately, a **common-rate binomial parametric bootstrap** (20,000 draws) tests whether the between-source spread exceeds a single-common-rate sampling model. Neither control identifies a database-construction mechanism. The per-source exclusivity rates also carry a bootstrap spread (10 resamples per source, reported in SI-H), so between-source differences are not read off single point estimates. Because the two databases define "a composition" with different keys and cover different row sets, no ordering of their magnitudes is claimed; this is a bounded support audit under a second set of metadata definitions, not an independent replication.

Under ARC-MOF's non-equivalent composition definitions, high coupling is observed after the stated singleton and permutation checks. Source dependence and identifier definitions limit the comparison.

3.6 Representation, novelty and external-feasibility checks

Adding linker identifiers to B reduced pooled LOTO MAE for both targets and both ensembles in all five 25,000-row redraws; the 100,000-row single-sample check preserved those directions (Table S19). This does not imply that I uniformly outperforms numerical descriptors: five-redraw mean uptake/RF MAE was 0.2510 for I versus 0.2400 for G, despite slightly higher mean R^2 for I. On the mapped subset, T increased pooled MAE relative to I in all five redraws for both targets and ensembles. That subset retains approximately 24% of valid uptake rows and eight of eleven topologies. Repeating the contrast with all features considered at each split (`max_features=1.0`, everything else fixed) leaves the uptake direction unchanged — T − I = +0.122 (RF) and +0.125 (ET) mmol/g, positive in all five redraws — but shrinks the log-selectivity gap to +0.001 (RF) and +0.011 (ET) (SI-I.7, Table S25). Parent mappings, population selection and the fixed model configuration limit this result; it does not establish a general disadvantage of molecular fingerprints.

Pooled directions hide per-topology reversals, so the same contrasts were resolved topology by topology (five-redraw mean of the paired difference; first representation minus second, negative favours the first). For uptake/RF on the full sample, I−B is negative in 7 of 11 topologies and positive in bcu, fof, tbo and the, with differences from −0.023 to +0.035 mmol/g. On the mapped subset, T−I is positive on average for uptake/RF yet negative in 3 of 8 topologies, and single-topology changes reach +0.277 mmol/g. A pooled mean is therefore a summary over opposing local effects, not a uniform improvement or degradation (Figure 2c,d); which topologies reverse is listed in SI-I.1. "Not uniformly beneficial" therefore means that the representation change **reverses direction from topology to topology** — the sign of I−B flips in 4 of the 11 topologies (named above) — not that one representation is globally worse than another.

Constituent-level novelty also changes the interpretation of exact-key loss (Table S20). In the 25,000-row seed-42 uptake audit, 52.9% of unseen-key predictions recombine identifiers already observed in training; the corresponding 100,000-row value is 52.1%. The constituent-unseen category is dominated by unseen database metal-node identifiers, which are not chemical elements. In the seed-42 I/RF uptake LOTO audit, errors do not increase monotonically from C0 to C3: C1 MAE exceeds C2 at both sample sizes. Numerical-support and fold-level diagnostics likewise fail to establish a calibrated error ordering (Table S22); complete key mismatch occurs across groups with widely differing MAE. These measures describe support, rather than supplying validated prediction-error warnings or exclusion thresholds for screening.

The CoRE pilot examines experimentally derived structures but does not independently validate the support claim (Table S21). Forty sampled CIFs were parsed; an automated deconstruction routine returned metal-containing and organic-fragment candidates with bond cuts for 38. None of the 40 sampled records has completed expert validation of its chemical assignment. A ten-structure topology pilot returned nine net keys and six named-net matches, also unvalidated. A database-wide comparable support audit and an external adsorption-prediction test have therefore not been completed.

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4. Discussion

4.1 What this means for the stress-test protocol

The support audit establishes what the split removes under the specified identifiers: 74.7% of coarse composition keys are single-topology and 61.3% of structures lose an exact-key counterpart. It does not equate key absence with absence of constituent building blocks or structurally similar chemistry. The 21.1% protocol ratio concerns different validation constructions under the original representation; it is not a fraction of error caused by chemistry. The +70.0% nbo-side intervention is a local model response on one pair. Keeping these three quantities separate prevents a support count from being mistaken for a causal error attribution.

Topology hold-out remains a useful combined stress test. Its score quantifies transfer to the held-out topology groups represented in the source library under a specified learner and representation. It does not by itself establish transfer to experimental materials or to arbitrary new chemical structures. Once validated chemistry and grouping identifiers exist, support counting is inexpensive; obtaining comparable identifiers from experimental crystal structures may require substantial curation, as the CoRE pilot illustrates.

4.2 Relation to prior work and mechanistic hypotheses

The contribution extends the combined-stress-test framing already used by Asgari et al. and Alimardani et al. (refs. 16, 8) by quantifying exact-key support loss and testing the limits of its performance interpretation. The original omitted-variable analysis remains descriptive: group-associated η^2 changes from 0.168 to 0.031 for topology and from 0.393 to 0.255 for linker pair after numerical-descriptor residualisation, and linker-pair labels predict residuals with $R^2 = 0.214$. Different residual and raw variance denominators prevent interpreting these changes as shares of variance absorbed.

The follow-up I−B comparison directly tests adding linker identifiers and shows lower MAE, but I−G and mapped T−I prevent treating that improvement as a general solution to topology transfer. Numerical descriptors already contain chemical information through quantities such as weight and volume; G is not a chemically orthogonal control. With a fixed feature-subsampling rule, changing feature dimension also changes the opportunities for a tree to select each descriptor. The comparisons therefore evaluate specified representations and learners, not isolated quantities of chemical information.

Similarly, the matched replacement and fixed-pool screening results do not support a general rule that same-key partner coverage improves prediction or ranking. The ten-target panel (SI-G) is descriptive because its targets are algebraically dependent; its ten outcomes are not ten independent replication units. The identity-availability and residual analyses (SI-E, SI-G) suggest hypotheses, but neither these associations nor the interventions identify an adsorption mechanism.

4.3 Three reporting principles

First, define the composition key, distinguish exact-key loss from unseen constituent identifiers, and report per-group support alongside grouped scores. State whether identities come from database labels, validated molecular structures or an approximate mapping. Support measures should not be presented as calibrated error-risk estimates without separate validation.

Second, hold the evaluation population fixed within a comparison and report both the aggregation rule and per-group performance. Here pooled R^2 of +0.3814 and macro R^2 of −0.5414 describe different weighting choices. For screening, state the budget and distinguish pooled out-of-fold rankings from the shortlist of an individual fitted model on a frozen candidate pool; report recovery and selected-property value together.

Third, disclose effective group counts, representation and learner settings, replication units, mapping coverage and source dependence. Sample redraws, learner seeds, repeated measurements on the same topologies and independent databases support different uncertainty claims. The 22-rule checklist in SI-A supplies a fuller reporting framework. Applying these principles can require additional curation or analysis where the necessary metadata or predictions are absent.

Specification. One table, one row per structure, four columns at minimum: the coarse key (`metal_linker`, `organic_linker1`, `organic_linker2`, with `functional_groups` added for the fine key) and the group label (topology). (i) Single-topology fraction = distinct keys occurring in exactly one group divided by all distinct keys, composition-weighted (74.7% in hMOF). (ii) Structure-level unseen fraction = rows whose key has no counterpart in the training groups after the hold-out divided by all rows, structure-weighted (61.3%). (iii) Entropy ratio = H(T | C)/H(T), structure-weighted plug-in estimate with the first-order Miller–Madow correction reported separately as a sensitivity diagnostic (0.180, i.e. 82.0% of topology entropy removed), with the reverse ratio H(C | T)/H(C) = 0.860 reported beside it; the composition-weighted forward value (11.4%) is a different quantity and is not interchangeable with 18.0%. Pairwise overlap = shared keys divided by the union (Jaccard) for every group pair. For a table with these fields, run `python scripts/confound_diagnostic.py --csv DATA.csv --group-col topology --coarse-cols metal_linker,organic_linker1,organic_linker2 --pair-report`; other column names can be supplied through the same options. The separate `python scripts/singleton_audit.py verify` command checks the archived hMOF and ARC-MOF summary tables and does not recompute an audit from raw input.

4.4 Limitations and testable designs

Four limits remain after the follow-ups. (i) The support count is established in hMOF and in partly dependent ARC-MOF under non-equivalent keys. The CoRE MOF pilot (ref. 5) has not yet produced independently validated, comparable chemical identities and topology assignments for a database-wide audit; it supplies no external adsorption-prediction test. (ii) Only pcu ↔ nbo provides a usable matched topology pair. Replacement and screening repeats keep the same pair and candidate populations, while the original five sample redraws remain within one library. None estimates variation across independent topology pairs or experimental libraries. (iii) Original removal arms alter several properties of the training pool. The stricter replacement holds size and partner fraction fixed but leaves functional-group, metal-node and numerical-distribution differences; its effects cannot be assigned uniquely to shared chemistry. (iv) Representation tests cover two tree ensembles and two targets with fixed settings, plus limited original gradient-boosting and ridge checks; the full-feature sensitivity covers both learners only on the mapped subset and only random forest on the full samples. They do not establish robustness across graph models, broad hyperparameter searches or transferable molecular representations. The mapped subset omits three topology test groups and most rows. The one 100,000-row representation sample is a size-sensitivity check, not external replication. Three design limits are not removed by the follow-ups: the protocol ratio compares different fold partitions and training populations and sizes, evaluated on the same pooled scoring rows; the support census is a leave-one-topology-out quantity, whereas the scored five-fold split can withhold several topology groups together; and the screening comparison pools predictions from different folds before ranking.

The larger pcu population and its greater diversity also remain alternative explanations for the original directional contrasts. Random thinning controls row count within a side but does not make the two directions exchangeable. The follow-up replacement reverses neither that structural asymmetry nor all chemical imbalances.

The next priority is to validate structure-derived chemical keys and topology assignments on experimental structures, quantify full-library coverage and then determine whether additional matched pairs and comparable adsorption labels exist. Multiple validated pairs would allow the original and replacement contrasts to be tested across pairs; an untouched external candidate pool would test transport of ranking performance. Fold-level diagnostics require validation across independent groups before calibration as risk predictors. Feasibility and cost depend on identifier quality, pair coverage and label availability, and cannot be inferred from the low cost of metadata counting.

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5. Conclusion

In hMOF, topology hold-out removes exact composition-key support as well as topology labels: 74.7% of coarse keys occur in one topology and 61.3% of structures lose their same-key training counterpart. ARC-MOF shows high coupling under its own metadata definitions, with source dependence limiting the comparison. Performance consequences are more conditional. The original one-pair interventions and out-of-fold rankings show substantial contrasts, but the six follow-ups — representation, constituent novelty, CoRE feasibility, fold-level diagnostics, matched replacement and fixed-pool screening (SI-I) — do not support a single chemical mechanism, a calibrated support-to-error rule or a uniform screening benefit from shared composition. The 100,000-row check provides within-database sample-size sensitivity, and the full-feature re-run preserves the uptake contrasts while substantially reducing the mapped log-selectivity gap; the CoRE pilot leaves experimental-library validation open. Topology hold-out should be retained as a combined stress test and reported with explicit keys, support coverage, representations, group aggregation and screening design. These disclosures make the validation scope assessable without treating the score as an isolated measure of topological or chemical transfer.

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7. Tables

Table 1. Validation protocols, two panels. **Panel A:** five schemes on 100,000 samples, pooled R^2 under 426 features and the matched 415-feature check. Every Panel A value is pooled over one five-fold cross-validation run on the same 100,000 rows at seed 42, with a separate 120-tree random forest fitted in each fold and no seed averaging or hyperparameter tuning; the 415-feature column removes the topology one-hot block from both arms. **Panel B:** leave-one-topology-out aggregates of the same 100,000-sample run, from one 120-tree random forest per held-out topology (eleven models, single run at seed 42); the 25,000-sample five-seed LOTO replication is Table S9.

Panel A — five schemes (pooled R^2).

Scheme	426 features	415 features (topology block removed)
random	0.7529	0.7449
chemistry key	0.6738	0.6674
linker pair	0.6606	—
metal	0.5270	—
topology (GroupKFold)	0.3786	0.3845

Panel B — leave-one-topology-out aggregates.

Aggregation	pooled	weighted	macro	median
LOTO R^2	+0.3814	−0.0589	−0.5414	+0.2782

Source data: `outputs/leakage_results_final.csv`, `outputs/leakage_results_chemistry.csv`, `outputs/leakage_results_notopo100k.csv`, `outputs/metrics_audit.csv`.

Table 2. Controlled-contrast MAE (mean ± SD over five outer seeds), nbo and pcu test sets. n_train, MAE and RMSE are all five-seed means; the (c) SD is the nested seed-level value.

Test	arm	n_train	MAE	RMSE
nbo	a (with partner)	42,381	0.1065 ± 0.0032	0.1352 ± 0.0077
nbo	b (partner removed)	10,656	0.1811 ± 0.0136	0.2405 ± 0.0136
nbo	c (size-matched)	10,656	0.1071 ± 0.0048	0.1375 ± 0.0109
nbo	e (shared-composition removed)	30,039	0.1326 ± 0.0082	0.1745 ± 0.0127
nbo	f (size-matched from e)	10,656	0.1250 ± 0.0073	0.1697 ± 0.0145
pcu	a (with partner)	18,275	0.2424 ± 0.0038	0.4030 ± 0.0021
pcu	b (partner removed)	10,656	0.2396 ± 0.0010	0.4024 ± 0.0030
pcu	c (size-matched)	10,656	0.2364 ± 0.0031	0.4002 ± 0.0044
pcu	e (shared-composition removed)	11,586	0.2293 ± 0.0020	0.3935 ± 0.0032
pcu	f (size-matched from e)	10,656	0.2304 ± 0.0045	0.3961 ± 0.0066

Note: The (c) arm is the nested size-matched draw (`diversity_control.py`; five outer seeds × five draws, SD across seed means). The seed-42 repeated-draw (c) is a distinct replication design in SI-C. *Source data:* `outputs/isoreticular_control_seeds.csv`, `outputs/composition_matched_control.csv`, `outputs/arm_e_partner_coverage.csv`.

Table 3. ARC-MOF support audit under its own chemistry definitions.

granularity	N	M	non-singleton	single-topology (all)	single-topology (non-singleton)
coarse	256,351	180,989	49,869	99.3%	97.3%
fine	214,543	177,971	26,380	99.7%	98.2%

Source data: `outputs/confound_arcmof.csv` (N, M and all-composition fractions; `topo_col = Crystalnet`) and `outputs/arcmof_non_singleton_summary.csv` (non-singleton counts and fractions); the singleton counts and entropy corrections behind §3.1 and §3.5 are in `outputs/singleton_audit.csv`. M counts distinct keys among the rows that carry them: 180,989 among 256,351 coarse-valid rows, and 177,971 among the 214,543 rows that also carry a functional-group key; the fine key is therefore computed on 41,808 fewer rows, which is why its M is lower. Source-removal and null tables are in SI-H.

Supporting Information

Data sources, input-validity checks and model settings; grouped-split reporting checklist; composition-support, entropy and overlap audits; intervention, aggregation and screening results; ARC-MOF source analyses; representation, constituent-novelty, CoRE feasibility, fold-diagnostic, partner-replacement and fixed-pool analyses; feature-subsampling sensitivities; and supplementary figures (PDF).

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9. Declarations

Data availability

Code and data. The reported analyses are archived with the source code and derived results in release v1.4 at the fixed version DOI <https://doi.org/10.5281/zenodo.23618746>; the source repository is <https://github.com/wang359826/topology-holdout-support-audit>. The release includes the original analyses, six follow-up experiments (SI-I), tie and candidate-bootstrap audits, and feature-subsampling sensitivity. The files supplied with this submission are a submission revision built upon v1.4, including text and diagnostic-tool corrections; these revisions are not yet archived as a new public release. The concept DOI <https://doi.org/10.5281/zenodo.23415179> links the version series, while <https://doi.org/10.5281/zenodo.23434268> identifies the earlier v1.3 release containing the original analyses only. `REPRODUCE.md` maps reported results to their producers and states verification limits; `outputs/MD5SUMS.txt` records the submitted output files. Input data are available from hMOF at <https://archive.materialscloud.org/records/nsp7v-2dk91> and ARC-MOF v6 at <https://doi.org/10.5281/zenodo.13891643>.

CoRE pilot inputs originate from the public CoRE MOF 2019 ASR v1.1.0 record <https://doi.org/10.5281/zenodo.3370144>. This work generates **no new experimental or simulation data**. **Neither input database is redistributed with this manuscript or with the source-code archive**; both are publicly available without registration and are fetched by the reader. The source-code archive's `DATA.md` gives the exact retrieval commands, the expected byte sizes and the MD5 of every file used, and records two known upstream failure modes so a reader is not misled by them. Licences are those stated on the two record pages. Archived analysis outputs and final figures are stored at the locations specified in `REPRODUCE.md`; historical and newly generated outputs have separately documented verification and checksum scopes, and the Supporting Information and `REPRODUCE.md` state what was verified end to end and what was not.

Code availability

Analysis scripts, environment specifications and data-retrieval instructions accompany the submission. The reproduction supplement has three distinct verification entry points. Level 1 (`level1_verify/run_level1.ps1`) checks the documented table and claim summaries against archived outputs, together with numbering and citations; it fits no model and reads no raw input data. Level 2 (`level2_predictions/verify_predictions.py`) recomputes selected intervention and random/topology screening metrics from the archived row-level predictions; it does not cover all representation, fixed-pool or LOTO results. Level 3 (`level3_rerun/run_subset.py`) refits a representative seed-42, 25,000-row random-split comparison of G/B/I representations from the fetched hMOF data; it does not reproduce the full study. `REPRODUCE.md` specifies the input checksums, dependencies, commands and coverage of each entry point. The original source pipeline also uses an older five-level numbering scheme; its historical timings and numerical comparisons apply only to the dated step lists documented in that archive and are distinct from the three entry points supplied here. Neither the archived-output checks nor a representative rerun establishes a fresh end-to-end reproduction of every reported analysis.

Author contributions (CRediT)

The sole author performed all roles: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing.

Conflicts of interest

There are no conflicts of interest to declare.

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AI usage disclosure. Large language models were used during the development and checking of analysis code and during manuscript drafting and language revision. The author reviewed the code, checked numerical statements against the archived outputs and producer scripts, and reviewed and edited AI-assisted text before inclusion. The author is responsible for the study design, analytical decisions, interpretation and final content.

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